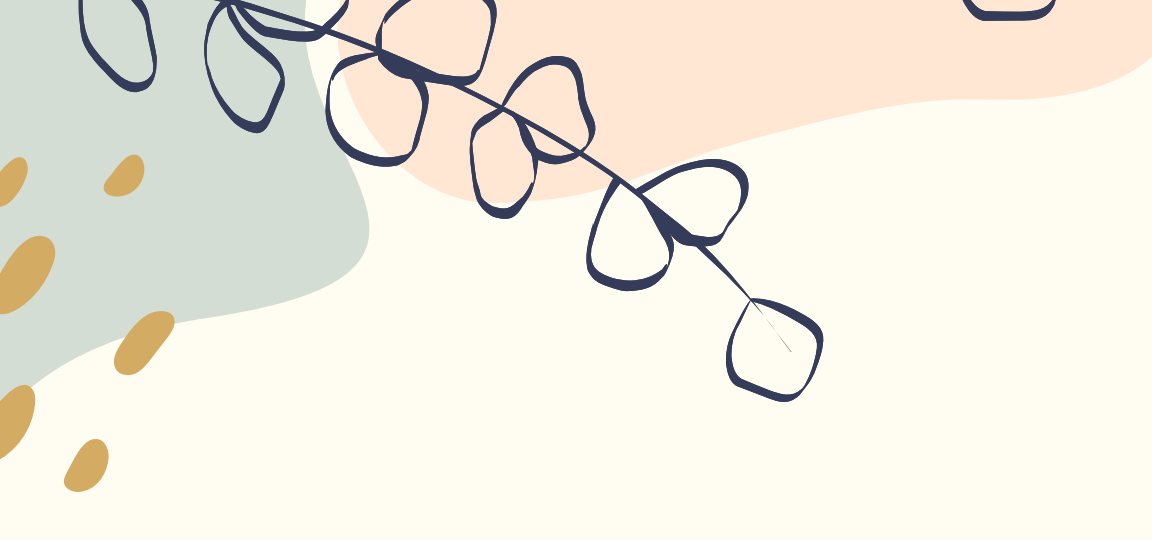




MENTAL HEALTH RISK PREDICTION

Can we predict mental health risk using lifestyle,
stress, and personal factors?

By Hayden Budhan | CSC 108 Angel Pineda



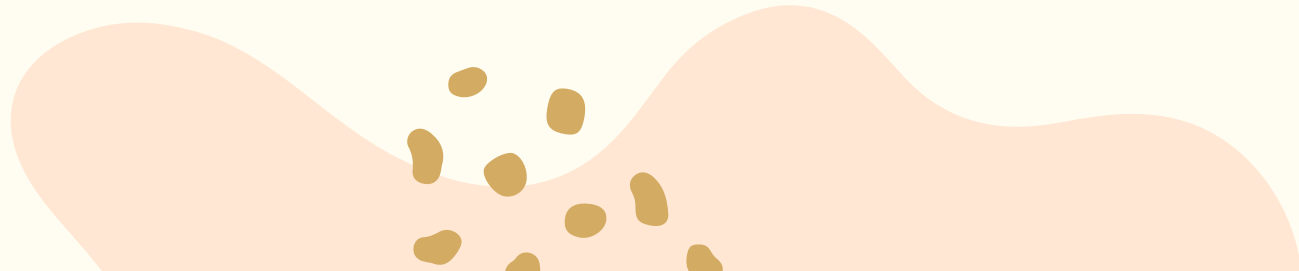
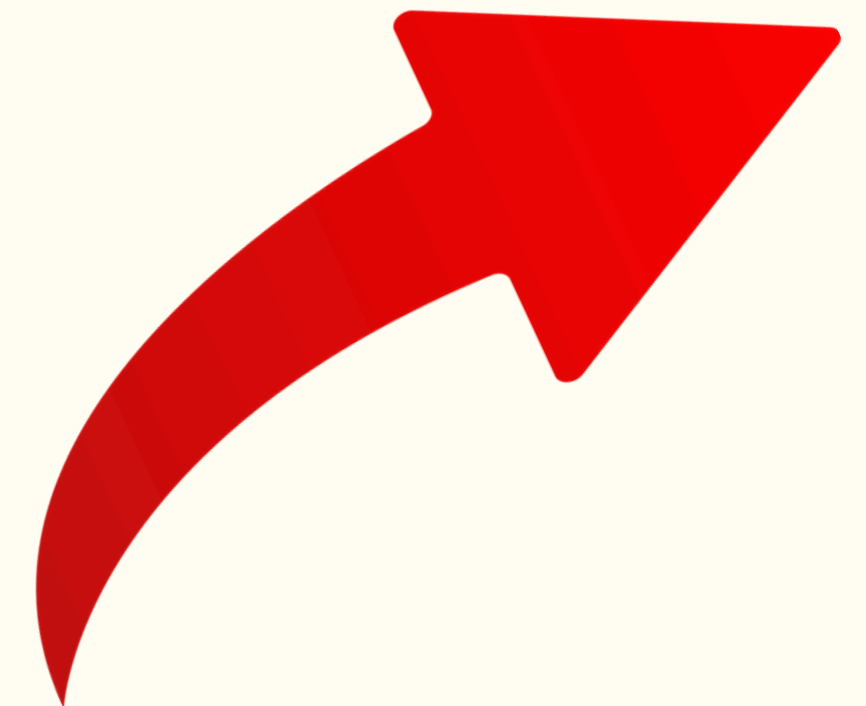
INTRODUCTION / PROBLEM

- **Can we predict mental health risk using lifestyle, stress, and personal factors?**
- Mental health is influenced by multiple variables such as sleep, stress, and emotional well being
- Goal: use data analysis and modeling to identify patterns and make predictions based on outside factors



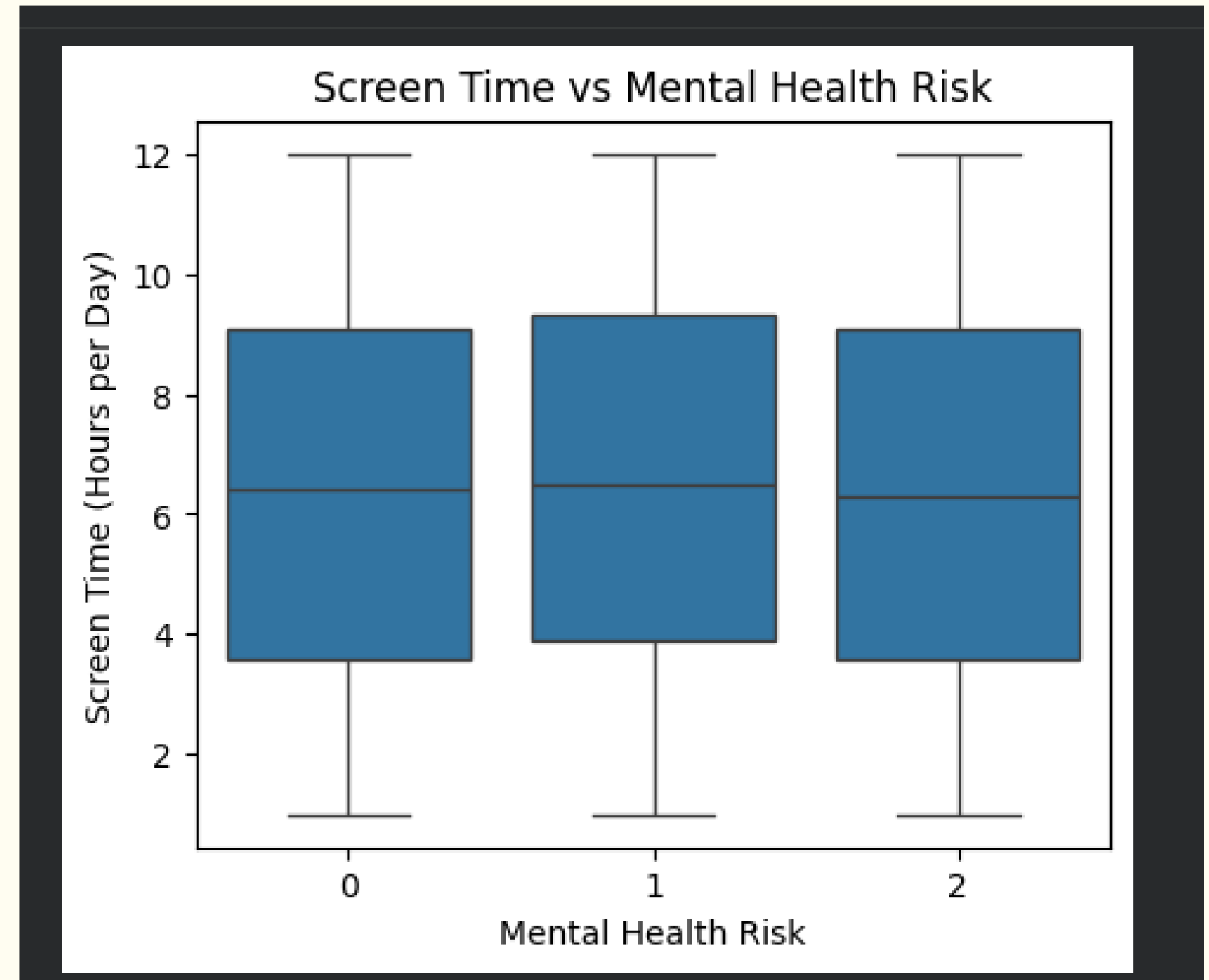
DATASET OVERVIEW

- Dataset includes lifestyle, stress, and psychological variables that may influence someone's mental health
- Key features used include sleep hours, anxiety score, depression score, and stress levels
- Target variable:
 - 0 = Low risk
 - 1 = Medium risk
 - 2 = High risk



SCREEN TIME AND MENTAL HEALTH

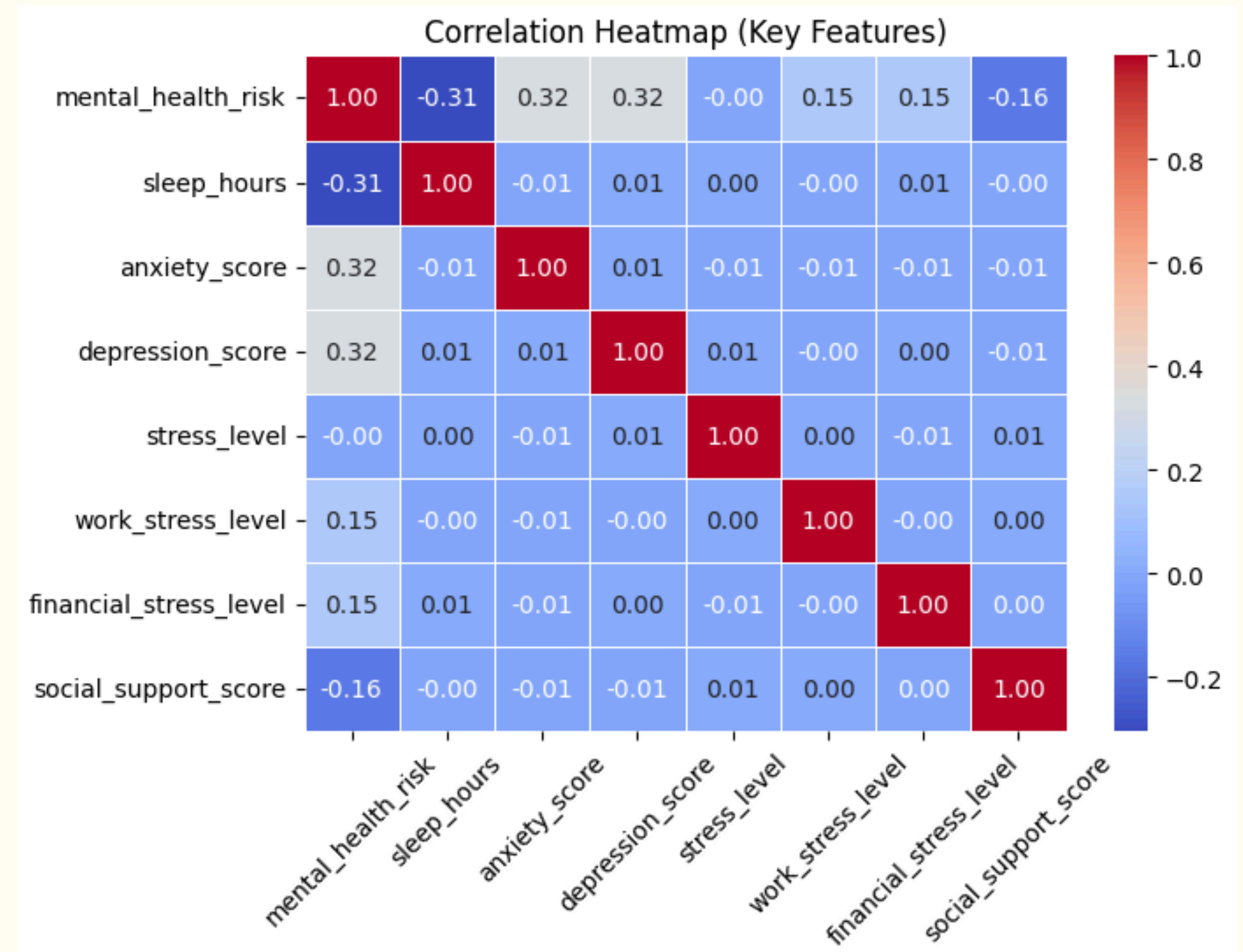
- Screen time was chosen due to its commonly assumed impact on mental health
- However, it shows little to no clear relationship with mental health risk
- Median and distribution are similar across all risk levels
- This suggests screen time is not a strong predictor of mental health risk



This graph shows that screen time is fairly similar across all risk levels, suggesting it may not be a strong factor in predicting mental health risk.

OTHER KEY PATTERNS

- Anxiety and depression show **strong positive relationships with mental health risk**
- In the heatmap, these appear as **darker red colors**, indicating **stronger positive correlations**
- Sleep shows a **negative relationship**, represented by **dark blue colors** in the heatmap
- Screen time shows **weak correlation**, with **lighter colors** indicating **little to no relationship**



BUILDING A PREDICTION MODEL

- A logistic regression model was used to predict mental health risk
- Input features included sleep, anxiety, depression, and stress related variables
- These features were selected based on earlier patterns and correlations
- The data was split into 80% training and 20% testing sets

```
from sklearn.model_selection import train_test_split # split data
from sklearn.linear_model import LogisticRegression # prediction model
from sklearn.metrics import accuracy_score # evaluation

features = [
    "sleep_hours",
    "anxiety_score",
    "depression_score",
    "work_stress_level",
    "financial_stress_level",
    "social_support_score"
]

X = mental_health_data[features] # input variables
y = mental_health_data["mental_health_risk"] # target variable

# We split data into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the model
model = LogisticRegression(max_iter=1000) # increase iterations to avoid errors

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

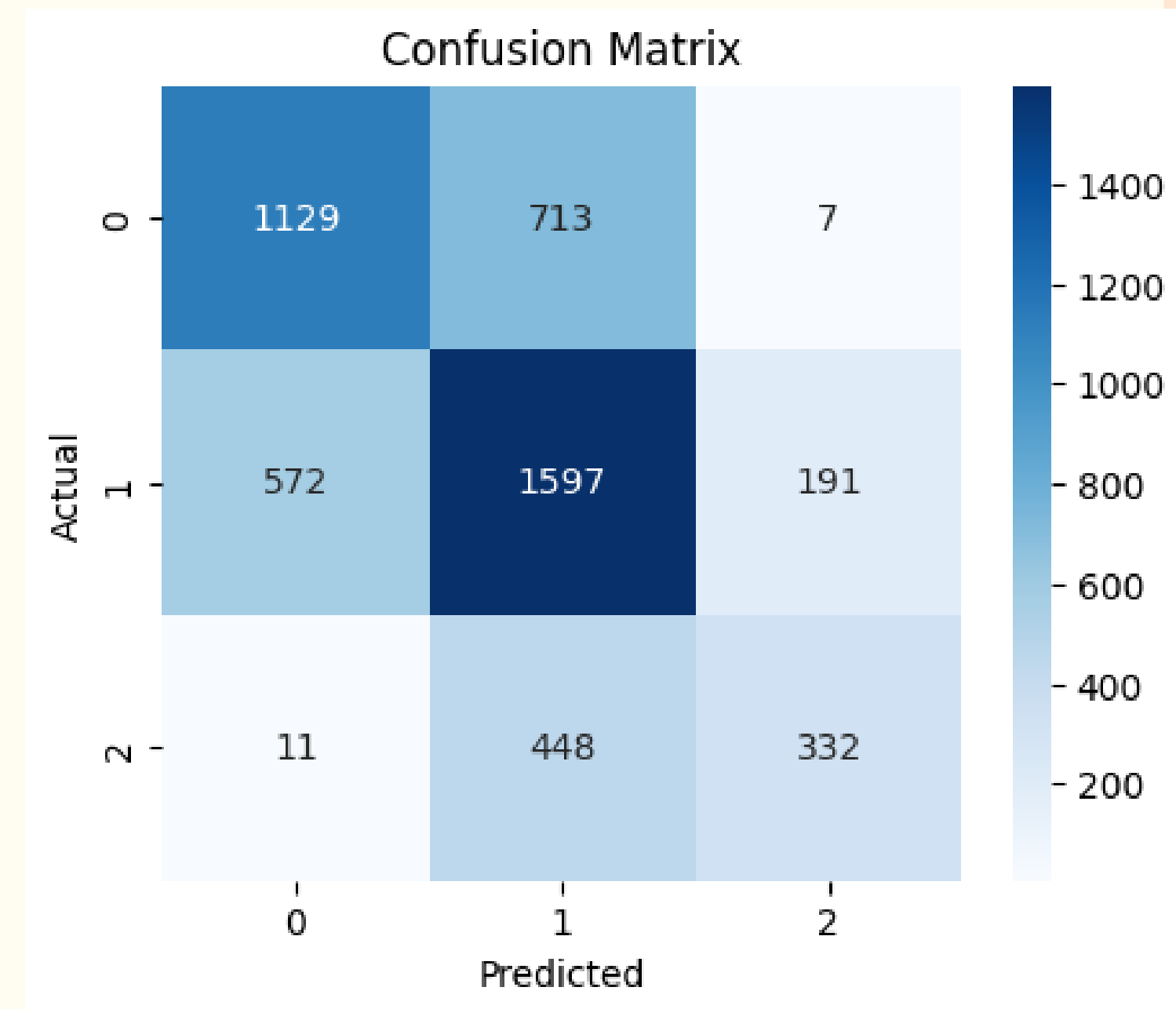
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", accuracy)
```

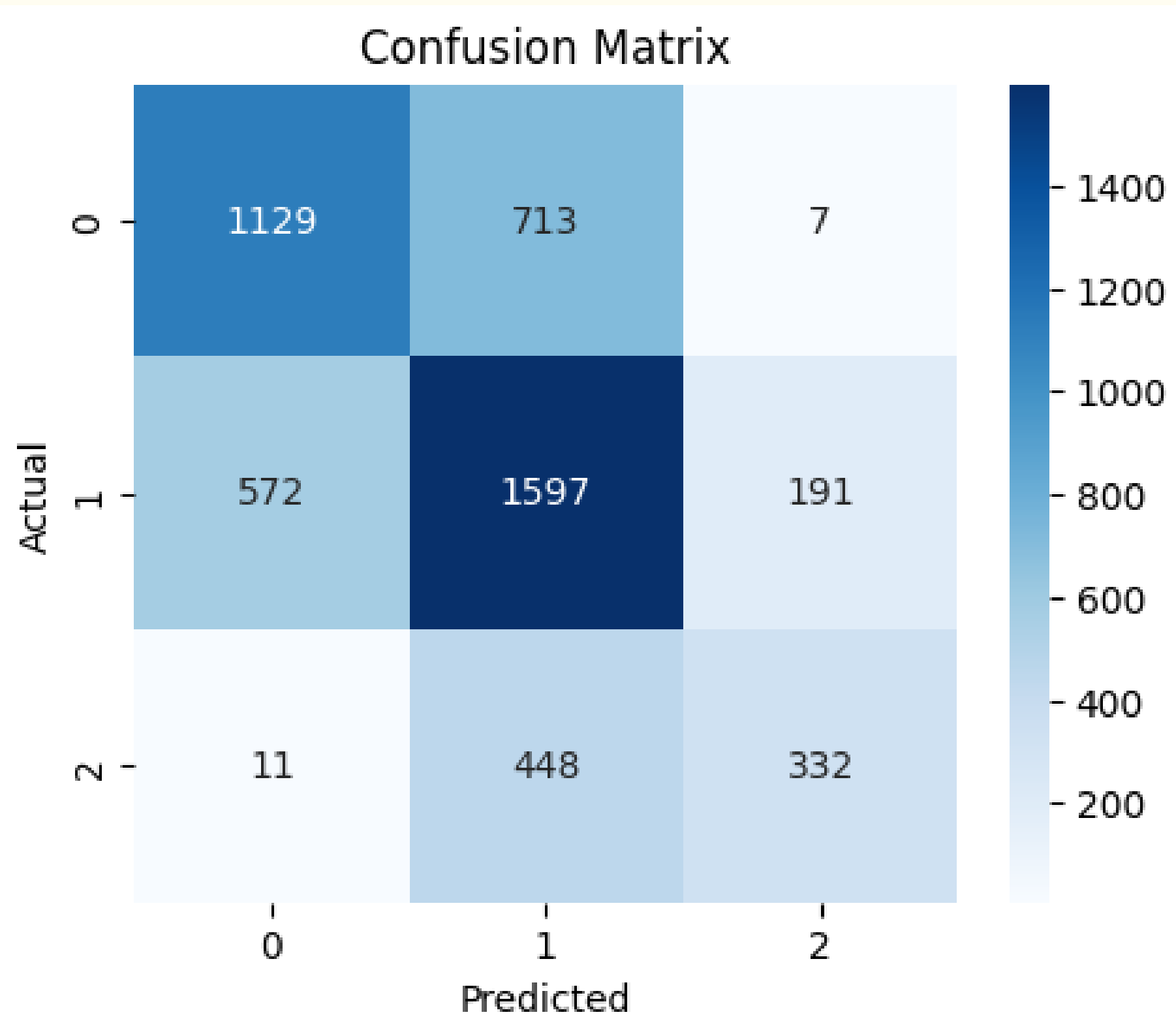
... Model Accuracy: 0.6116

MODEL PERFORMANCE

- Looking at a Confusion Matrix, the model achieved an accuracy of **approximately 61%**, which is more reliable than random guessing
- This indicates that mental health risk **can be predicted to some extent using the selected features**
- However, the accuracy is moderate, suggesting the problem is complex and not perfectly predictable



MODEL BEHAVIOR AND ERRORS



- The model performs best when predicting **medium mental health risk (1)**
- It frequently **misclassifies low (0) and high (2) risk as medium (1)**
- This indicates the model struggles to clearly distinguish **between extreme risk levels**
- The model tends to favor predicting the middle category

FEATURE IMPORTANCE (KEY PREDICTORS)

- **Sleep hours and social support** show **positive contributions**, associated with **lower mental health risk**
- **Anxiety and depression** have the **strongest negative contributions**, indicating **higher mental health risk**
- **Stress related variables** contribute, but have a **smaller impact compared to psychological factors**
- Overall, **sleep and psychological factors** are the **most influential predictors**

- Positive: Increases likelihood of lower risk. Negative: Increases likelihood of higher risk.

```
# Get feature importance (coefficients)
importance = model.coef_

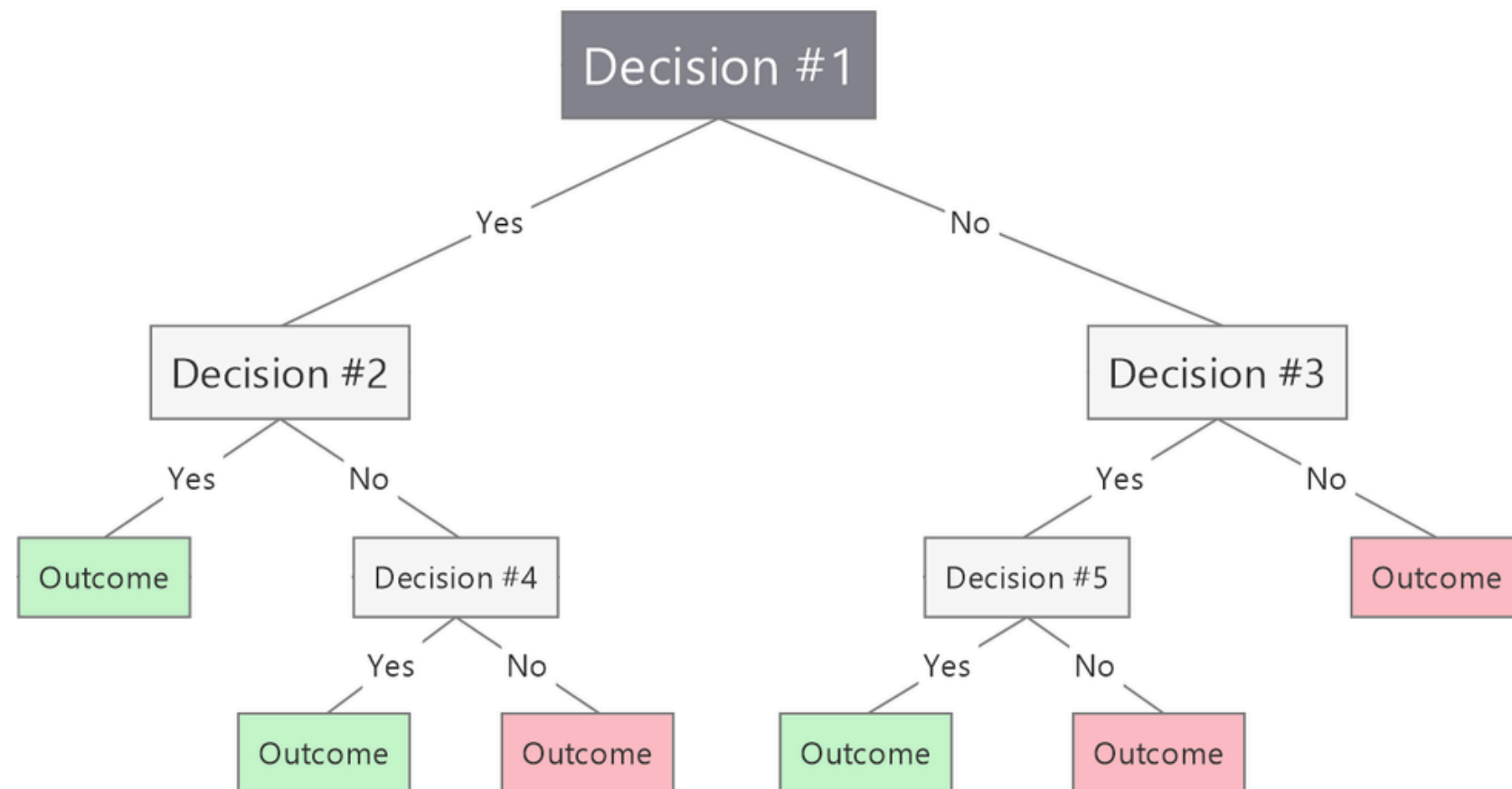
# Create DataFrame for readability
importance_df = pd.DataFrame({
    "Feature": features,
    "Importance": importance[0] # use first row for simplicity
})

# Sort values
importance_df = importance_df.sort_values(by="Importance", ascending=True)

print(importance_df)
```

	Feature	Importance
0	sleep_hours	0.351293
5	social_support_score	0.123021
4	financial_stress_level	-0.124249
3	work_stress_level	-0.124403
2	depression_score	-0.259633
1	anxiety_score	-0.259979

COMPARING PREDICTION MODELS

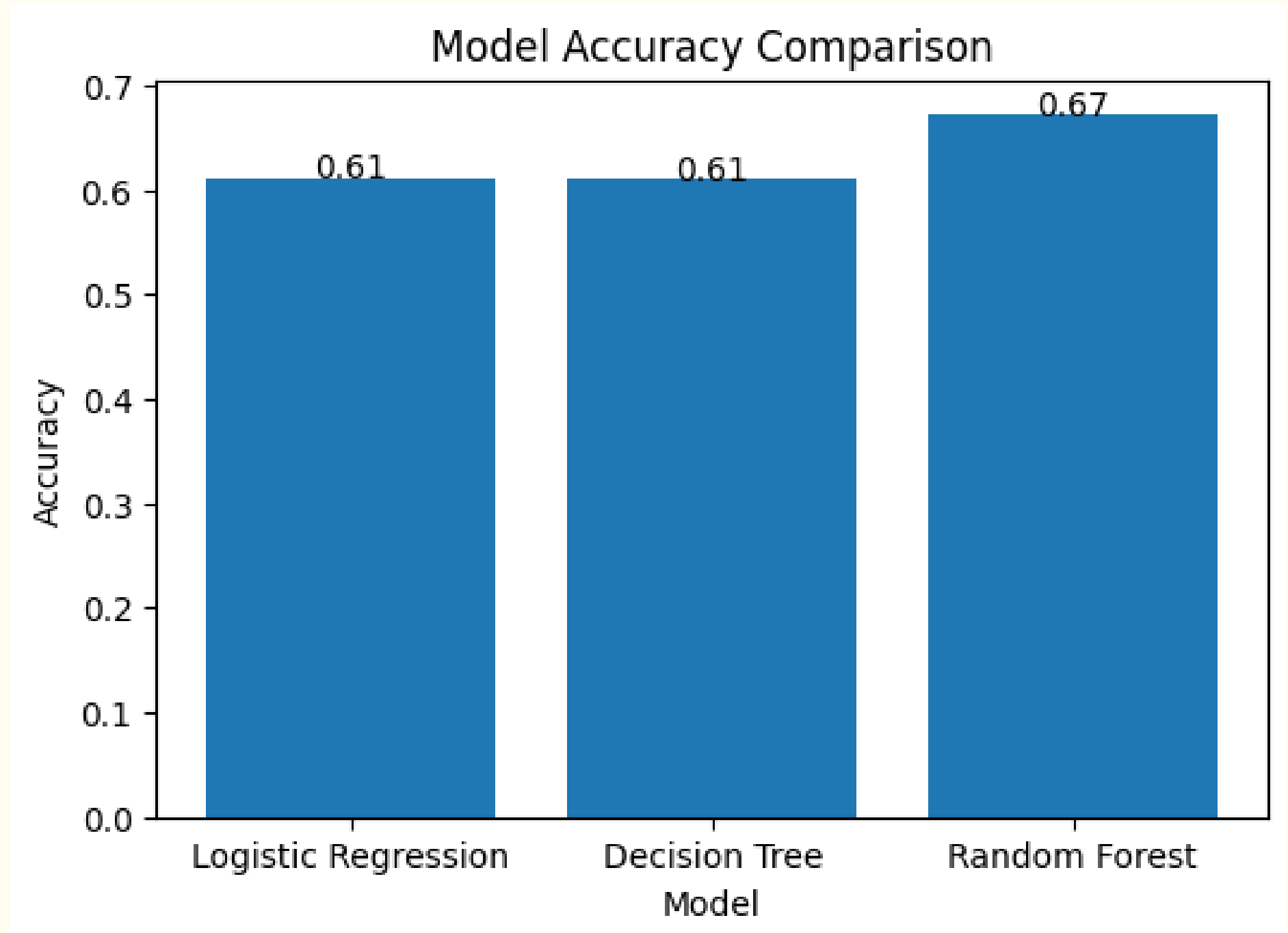


- In addition to logistic regression, other models were tested
- A **decision tree model** was implemented for comparison
- A **random forest model** was also used to **improve prediction performance**
- Comparing multiple models helps determine **which approach works best for this dataset**

MODEL ACCURACY COMPARISON

- **Logistic Regression Accuracy: ~61%**
- **Decision Tree Accuracy: ~61%**
- **Random Forest Accuracy: ~67%**
- **Random forest achieved the highest accuracy among the models**
- **This suggests more complex models can better capture relationships in the data**

- Positive: Increases likelihood of lower risk. Negative: Increases likelihood of higher risk.



KEY INSIGHTS AND CONCLUSION

- Mental health risk **can be predicted** using **lifestyle, stress, and psychological factors**, but with **moderate accuracy**
- **Anxiety, depression, and sleep** are the most **influential variables** in predicting mental health risk
- **Screen time** shows **little to no relationship**, suggesting it is not a strong predictor
- **Comparing models** showed that **random forest performed best**, indicating **more complex models improve prediction**
- Mental health is complex and influenced by additional factors not captured in the dataset (e.g., personal experiences, biological and social factors)